Venous photoplethysmography: Relationship between transducer position and regional distribution of venous insufficiency

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This study was performed to evaluate the photophlethysmographic technique in venous disease, especially regarding the effect of different positions of the photoplethysmographic transducer and the possibility of measuring regional changes in venous circulation. For this purpose photoplethysmographic recovery times were assessed from three different parts of the lower leg, that is, anterior midcalf level, medial midcalf level, and medial malleolar level. In patients with documented venous disease, regional photoplethysmographic findings were compared with results from venous pressure measurements, ascending/descending phlebography, foot volumetry, venous plethysmography, and Doppler investigation of the anterior tibial veins. The best correlation with pressure and volume recordings at the foot was obtained at the medial malleolar level. Only photoplethysmographic recovery times recorded at the anterior aspect of the leg were influenced by reflux in the anterior tibial veins. The present results indicate that photoplethysmographic recovery times reflect regional venous hemodynamics rather than overall venous hemodynamics in the limb. Photoplethysmographic recovery times were also found to be more related to superficial than to deep venous insufficiency. The results might have clinical implications when evaluating patients with regional venous dysfunction and in follow-up studies after venous surgery. (J VASC SURG 1990;11:436-40.)

Photoplethysmography (PPG) is widely used as a simple clinical method for assessing venous function noninvasively. The technique measures refill time of cutaneous blood content after exercise. A short refill time indicates venous reflux. The PPG transducer is most often applied at the anteromedial aspect of the distal lower leg,[1-3] even though other positions have occasionally been used.[4]

In a previous report from our department concerning venous circulation in patients who have undergone fasciotomy of the lower leg, we found that PPG mainly reflected regional changes in venous circulation, and that the position of the transducer was of crucial importance.[5]

The purpose of the present study was to assess whether the same relationship could be found in patients with venous insufficiency and to what extent the positioning of the PPG transducer, in relation to years). They were classified according to results from the regional distribution of the insufficiency, influenced the results obtained with PPG. For this reason PPG tracings were obtained from three different parts of the lower leg in patients with venous insufficiency, and the results were correlated with different noninvasive and invasive reference methods. These included ascending and descending phlebography, venous pressure measurements, and Doppler recordings used to characterize the degree and regional distribution of the venous disease.

Since the variations in PPG recovery times at different locations are not known, PPG recovery times were measured in a control group, from the same three different parts of the lower leg. Reproducibility and the effect of different types of exercise were also evaluated for regional PPG recovery times in this group.

MATERIAL AND METHODS

Controls. Photoplethysmography was performed on 16 legs in eight normal volunteers, four men and four women aged 21 to 37 years (mean 28 years). They were all without clinical evidence of venous insufficiency.

Patients. The patient group included 23 legs in 17 patients with chronic venous insufficiency, six men and 11 women aged 38 to 64 years (mean 53

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years). They were classified according to results from foot volumetry, venous plethysmography, Doppler ultrasonography, and ascending/descending phlebography. Most of the patients had a combination of deep and superficial venous insufficiency, in most cases they also had incompetent perforating veins. Three patients had deep venous reflux to the level of the calf measured with descending phlebography or venous plethysmography or both. However, results from foot volumetry in these patients were normal, indicating proximal deep venous disease with competent distal valves. Only three patients had isolated superficial venous disease with incompetence of the great saphenous vein. No patients with phlebographic or plethysmographic evidence of venous obstruction were included.

Photoplethysmography (PPG). A photoplethysmograph (EC-4 Plethysmograph, D.E. Hokanson Inc., Issaquah, Wash.) operated in DC mode was used to record changes in cutaneous blood content.[1,2] Three different sites for the PPG transducer were used—anterior midcalf level, medial midcalf level, and medial malleolar level. The instrumentation allowed recording from two sites simultaneously. Recordings were made with the subject sitting with the feet unloaded. The cutaneous blood content was recorded before, during, and after five dorsiflexions of the foot, once every other second. Venous recovery time (VRT) and half refilling time (T½) were calculated from the mean of two tests. Venous recovery time was measured from the end of exercise to the point at which a horizontal baseline was reached for 5 seconds. In the patient group, the study was repeated after occlusion of superficial veins by a below-knee cuff inflated to 80 mm Hg.

In the control group, the effect of 10 dorsiflexions compared with five was studied in 12 of the legs, and the reproducibility of each test was evaluated. For this, the test error was calculated from consecutive readings as well as from repeated studies over periods from 1 day to 3 months.

Noninvasive reference methods

Foot volumetry. Foot volumetry was performed as described by Norgren.[6] The following parameters were calculated with and without occlusion of superficial veins below the knee: expelled volume, relative expelled volume, refilling flow, T½, and refilling flow in relation to relative expelled volume.

Venous plethysmography. Venous plethysmography was done essentially as described by Bygdeman et al.[7] Strain-gauge wires applied around the thickest part of the calf were used to evaluate rate of venous emptying and to measure volume changes with tilting from supine to nearly erect with and without occlusion of superficial veins below the knee. Reference values established earlier were used both for foot volumetry[6] and venous plethysmography.[7]

Doppler ultrasound. Regional venous circulation at the anterior aspect of the leg was evaluated in 17 of the legs in the patient group. In these patients flow patterns were assessed in the anterior tibial veins by use of a 10 MHz bidirectional continuous wave Doppler probe. Venous flow was identified close to the anterior tibial artery where the artery becomes superficial at the anterior aspect of the lower leg. Bidirectional venous flow with the patient in sitting position, after manual compression of the foot, was used for diagnosing incompetence of the anterior tibial veins.[8]

Invasive reference methods

Phlebography. Eighteen of the 23 legs in the patient group were investigated with ascending and descending phlebography. Postphlebitic changes seen on ascending phlebography were used to denote deep venous disease. If possible, incompetent perforating veins were identified. Descending phlebography was performed with the patient supine and with evaluation of venous valve function during a sustained Valsalva maneuver.[9]

Venous pressure. By use of the same cannula as for ascending phlebography, venous pressure was measured in a distal dorsal foot vein via a standard pressure transducer. Venous recovery time and T½ were calculated after five kneebends, once every other second. Pressure measurements were repeated after occlusion of superficial veins by a below-knee cuff inflated to 80 mm Hg. In general, the invasive studies were performed the day after the noninvasive studies.

Statistics. Data are expressed as mean ± SEM. Differences of means were tested for statistical significance with Student's t test for paired and unpaired data. Linear regression was used for analyzing association between PPG results and results from venous pressure measurements, foot volumetry, and venous plethysmography.

Statistical significance was set at $p < 0.05$. Reproducibility was expressed as methodologic error (E) calculated from duplicate determinations according to the following formula: E = SD of the difference × 100/total mean × $\sqrt{2}$.

RESULTS

Venous recovery times in the control group are shown in Table I. Recovery times were significantly

Table I. PPG recovery times (mean ± SEM)

	VRT_5	VRT_{10}	
Anterior PPG	35 ± 5.5 seconds	41 ± 6.9 seconds	$p < 0.05$
Medial PPG	40 ± 5.2 seconds	47 ± 5.4 seconds	$p < 0.05$
Malleolar PPG	45 ± 5.7 seconds	65 ± 8.4 seconds	$p < 0.001$

Recovery times in healthy limbs (n = 16) after five (VRT_5) and 10 (VRT_{10}) dorsiflexions of the foot with the transducer at the anterior midcalf level, medial midcalf level, and medial malleolar level.

Table II. Reproducibility of PPG in healthy limbs (n = 12)

	VRT_5		VRT_{10}	
	E_1	E_2	E_1	E_2
Anterior PPG	19%	52%	10%	29%
Medial PPG	10%	10%	12%	16%
Malleolar PPG	11%	20%	6%	19%

Calculated errors from consecutive (= E_1) and repeated studies (= E_2). For abbreviations, see Table I.

longer after 10 dorsiflexions than after five at all three PPG transducer positions. After 10 dorsiflexions, VRTs were significantly shorter at the anterior and medial midcalf level than at the malleolar position ($p < 0.01$ and $p < 0.05$, respectively). Calculated errors were approximately 10% when comparing two consecutive measurements and approximately 20% when comparing initial and repeated studies. Measurements from the anterior aspect of the leg gave somewhat higher errors (Table II).

Photoplethysmographic results in the patient group are shown in Fig. 1. At all measuring points, VRTs were significantly shorter in the diseased legs than in the control legs (anterior midcalf $p < 0.01$, medial midcalf and medial malleolar $p < 0.0001$). Linear regression analyses demonstrated that VRT and T½ obtained from the malleolar PPG transducer site correlated best with those obtained with venous pressure measurements and foot volumetry. However, this correlation was strongly reduced when recovery times were compared after occlusion of superficial veins (Table III). No correlation was found between PPG results and venous reflux expressed as volume increase at tilting from supine to nearly standing position measured with venous plethysmography with strain-gauge wires at midcalf level.

With regard to the three patients with proximal deep venous insufficiency and competent distal valves, different photoplethysmographic patterns could be observed. In one of these patients PPG indicated only superficial venous disease at all three measuring points. Another had PPG recovery times around 30 seconds at midcalf level and 69 seconds at malleolar level. The third patient showed recovery times between 40 and 45 seconds at all three measuring points. This patient had no evidence of concomitant superficial venous disease according to reference tests.

For analyzing regional venous circulation at the anterior aspect of the leg, the patient group was subdivided, according to Doppler results, into patients with anterior tibial vein reflux (n = 8) and patients without reflux (n = 9). Photoplethysmographic recovery time at the anterior aspect of the leg was 13 ± 1.0 seconds in the reflux group and 22 ± 6.9 seconds in the group without reflux. In the reflux group only, this value was significantly shorter than VRT obtained from the anterior aspect of the leg in the control legs ($p < 0.01$). This difference was not found with the PPG transducer at the medial midcalf or malleolar position, which showed significantly shorter recovery times than control legs did, regardless of whether the anterior tibial veins were incompetent or not.

DISCUSSION

In previous studies with PPG various durations and types of exercise are used, for example, plantarflexions/dorsiflexions of the foot,[1,3,10,11] heel-raisings,[12] kneebends,[13] and treadmill walking.[14] The present results from PPG tracings in the control group showed that PPG recovery times varied with the length of exercise performed and with transducer positioning. This underlines the need for each laboratory to obtain its own PPG reference values for its test model.

The present study showed relatively high variability when repeated PPG recordings were made from healthy controls. This was most probably due to variations in the rate of arterial inflow. For follow-up studies, a correction for arterial inflow rate might improve the usefulness of PPG,[15] although in patients with venous reflux, variations of arterial inflow rate will be of less quantitative importance.

The present results indicate that a short PPG recovery time reflects regional venous dysfunction rather than overall venous hemodynamics in the

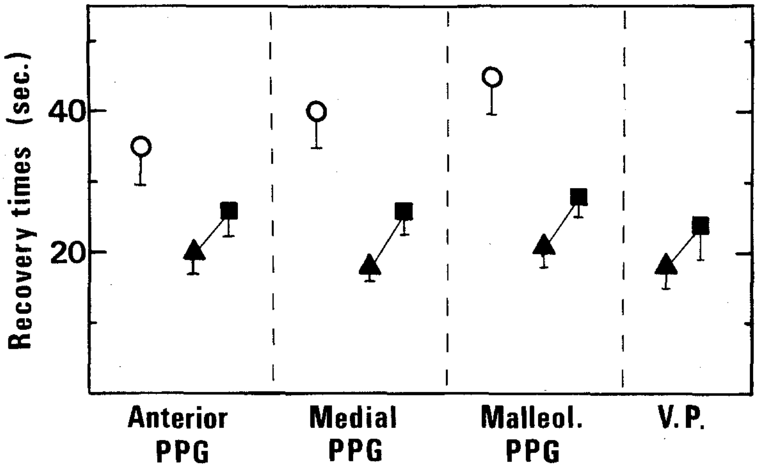

Fig. 1. Venous recovery times (mean ± SEM) in patients with venous disease, obtained with PPG (anterior midcalf level, medial midcalf level, and medial malleolar level, n = 23) and intravenous pressure measurements (V.P., n = 18) without (▲) and with (■) below-knee tourniquet. ○ = control legs (n = 16).

Table III. Correlation between VRT assessed with venous pressure measurement and PPG and T½ assessed with foot volumetry and PPG, without and with superficial venous occlusion

	Correlation coefficient (r)			
	Venous pressure (VRT)		*Foot volumetry (T½)*	
Anterior PPG	0.74	($p < 0.001$)	0.54	($p < 0.01$)
with occlusion	0.44	(NS)	0.38	(NS)
Medial PPG	0.69	($p < 0.01$)	0.39	(NS)
with occlusion	0.57	($p < 0.05$)	0.25	(NS)
Malleolar PPG	0.81	($p < 0.001$)	0.84	($p < 0.001$)
with occlusion	0.44	(NS)	0.54	($p < 0.05$)

lower limb. Results from Doppler investigation and volume/pressure recordings both support this view. Only VRTs obtained from the anterior aspect of the lower leg were influenced by reflux in the anterior tibial veins. Moreover, when PPG findings were related to volume and pressure changes at the foot, high correlation coefficients were obtained only with PPG at the malleolar level, that is, the level closest to the foot. Thus, only PPG at the malleolar level reflected distal valve function accurately. However, in patients with venous disease but with competent distal valves, other positions of the PPG transducer might be more useful for characterizing the degree and regional distribution of the venous disease.

Photoplethysmography and foot volumetry have been used together in studies of venous hemodynamics.[16] However, no previous study has correlated recovery times obtained with these two methods. In the present study a close correlation was found between T½ assessed with PPG at the malleolar level and T½ assessed with foot volumetry.

Excellent correlations between simultaneously recorded recovery times by PPG at the malleolar level and venous pressure are reported in previous studies,[1,10,14] mainly, however, by recordings without superficial venous occlusion. The present results are in line with these findings, even if somewhat lower correlation coefficients were obtained with nonsimultaneous recordings. However, after occlusion of superficial veins below the knee, no correlation between PPG recovery time at the malleolar level and venous pressure measurement was shown in the actual study. This indicates that PPG recovery times are more related to superficial than to deep venous insufficiency. A previous report by Barnes and Yao[12] supports this finding. They stated that PPG mainly documents reflux in perforating or superficial veins. However, deep venous reflux in the anterior tibial veins was, in the present study, found to influence PPG recovery times from the anterior aspect of the leg. This was probably due to the fact that superficial venous disease is relatively uncommon on the anterior aspect of the leg.

Resional PPG tracings are, of course, of minor importance in patients with total venous insufficiency. In advanced venous disease, PPG recovery times could be said to represent overall hemodynamics in the limb. However, when evaluating various interventions, the positioning of the transducer might be more critical. The present results with PPG as a reflector of regional venous circulation could be the explanation of why some authors, in spite of good clinical results, have failed to document any hemodynamic improvement after proximal deep venous surgery[17,18] or minor local surgical procedures.[10]

In conclusion, the present results with PPG tracings obtained from different parts of the lower leg indicate that PPG recovery times reflect mainly regional venous hemodynamics. Photoplethysmographic recovery times were also found to be more related to superficial than to deep venous insufficiency. The results might have clinical implications when evaluating patients with regional venous dysfunction and in follow-up studies after venous surgery.

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